

Physics Exam 07 — Ch 1-6+9 (Full-Year Cumulative) | Marking Report

Candidate: Seemab | Date: 25 June 2026 | Exam: exam-07-ch1-6-9 | Federal Board Annual 2026 Format | Syllabus: Physics Ch 1, 2, 3, 4, 5, 6, 9

Correction note (2026-06-25 deep re-check): Q.3(iii)OR re-marked from 3/5 to 2/5. Original marking incorrectly deducted only 0.5 for the absent Hooke's law applications (a 2-mark sub-part scoring 0) while over-deducting 1.0 for "no sketch" (correct deduction is 0.5 since all three required labels were described correctly). Corrected breakdown: Elasticity 0.5, Hooke's law 1.0, Graph 0.5, Applications 0.0 = 2/5. Section C total corrected from 16/20 to 15/20. Grand total corrected from 59.5/65 (91.5%) to **58.5/65 (90.0%, Grade A+)**.

58.5/65

GRAND TOTAL

A+

GRADE (90.0%)

12/12

SECTION A — MCQS

31.5/33

SECTION B — SHORT ANSWERS

15/20

SECTION C — LONG ANSWERS

Section A — Multiple Choice Questions (12/12)

Q	Seemab	Key	Result	Q	Seemab	Key	Result
1	B	B	✓	7	C	C	✓
2	B	B	✓	8	C	C	✓
3	C	C	✓	9	C	C	✓
4	B	B	✓	10	C	C	✓
5	B	B	✓	11	B	B	✓
6	B	B	✓	12	D	D	✓

Perfect MCQ score — 12/12. 6th consecutive perfect MCQ result (exams 14-19). Physics MCQ record: 60/60 all-time across 5 Physics exams.

Section B — Short Answers (31.5/33)

Q.2(i) — Types of Motion

3/3

All three types defined correctly (translatory, rotatory, vibratory) with distinguishing criteria. Full marks.

Q.2(ii) — Distance and Displacement

2.5/3

Displacement defined correctly (shortest path between two points, vector). Distance slip: written as "longest path" instead of "total path length" (scalar, all points covered). The word "longest" implies a specific route choice rather than the actual route taken.

- −0.5: distance defined as "longest path" rather than "total path length"

Q.2(iii) — Newton's Laws

3/3

All three Newton's laws stated with full mathematical expressions ($F = ma$, $F_{ae} = -F_{sm}$) and clear everyday examples. Precise textbook language throughout. Full marks.

Q.2(iv) — Torque

3/3

Torque defined as $\tau = r \times F$ with units N·m, moment arm explained, and numerical calculation correct. Full marks.

Q.2(v) — Work and Energy

3/3

$W = Fd \cos\theta$ stated and applied correctly. Kinetic energy formula and potential energy formula both given with examples. Full marks.

Q.2(vi) OR — Advantages and Disadvantages of Friction

2.5/3

Advantages section complete (walking, brakes, grip). Disadvantages section: only one disadvantage written (heat generation / energy loss). The question required two disadvantages.

- −0.5: only one disadvantage of friction given; second disadvantage (e.g., wear and tear of surfaces / noise) required

Q.2(vii) — Power

3/3

$P = W/t$ defined with SI unit (Watt), converted to horsepower correctly, numerical application complete. Full marks.

Q.2(viii) OR — Liquid Pressure ($P = \rho gh$ Derivation)

2.5/3

Full derivation present: column of liquid, $F = mg$, $m = \rho V$, $V = Ah$, therefore $P = \rho gh$ — all steps shown. The conclusion sentence explicitly stating that pressure increases with depth was absent.

- −0.5: derivation complete but the concluding statement (pressure depends on depth h , not on area or shape of container) not written

Q.2(ix) — Archimedes' Principle / Buoyancy

3/3

Statement, formula, and application (ships, hydrometers) all correct. Full marks.

Q.2(x) — Conservation of Energy

3/3

Law stated precisely. Pendulum example with full KE-PE interchange at each point. Full marks.

Q.2(xi) — Nature of Physics / Scientific Method

3/3

Physics definition, distinction of hypothesis / theory / law, and comparison with technology and engineering — all complete and accurate. Full marks.

Section C — Long Answers (15/20)

Q.3(i) OR — Types of Motion with Daily-Life Examples

4/5

All three types defined clearly. Daily-life examples provided: rectilinear/translatory — a car on a straight road (correct); rotatory — Earth rotating on its own axis AND fan blades (two examples, both correct). Vibratory motion: defined correctly but no daily-life example given (e.g., a swinging pendulum, guitar string, vibrating tuning fork). The single deduction is solely for the absent vibratory-motion example.

- -1: no daily-life example given for vibratory motion (a pendulum, guitar string, or vibrating tuning fork would have earned the mark)

Q.3(ii) — Newton's Laws (Extended)

4/5

All three laws written with full mathematical statements. Third law specifically shows $F_{as} = -F_{sa}$ notation correctly. Worked examples for each law. The "limitations of Newton's laws" sub-part (Newton's laws do not apply at very high speeds approaching c , or at atomic/sub-atomic scales) was entirely absent.

- -1: limitations of Newton's laws sub-part entirely absent

Q.3(iii) OR — Elasticity, Hooke's Law, Graph, Applications

2/5

Mark-by-sub-part breakdown (answer key allocation: Elasticity 1, Hooke's law 1, Graph 1, Three applications 2):

- **Elasticity — 0.5/1:** First sentence correct (property of a material to regain original shape after removing force). Second sentence in error: "materials that do not recover is called elastic limit" — the correct term for a material that does not recover is *inelastic*; elastic limit is the stress point beyond which permanent deformation occurs. -0.5
- **Hooke's law — 1/1:** $F = kx$ stated, k defined as spring constant with units N/m, formula applied correctly. Full credit.
- **Graph — 0.5/1:** All three required label points correctly identified: limit of proportionality (A), elastic limit (B), breaking point (D). The graph was described in words with labels referenced. However, no actual sketch of the force-extension curve was drawn on the answer sheet. The labels alone without a drawn curve earn half credit. -0.5 for missing sketch drawing
- **Three practical applications — 0/2:** Entirely absent. The answer moves directly to Q.3(iv) without listing any applications. The expected answers are: spring balance (measures force/weight using $F = kx$), watch balance wheel (controls oscillation frequency), galvanometer (current measured via spring deflection). A 2-mark sub-part scoring 0 is the primary driver of this question's low mark. -2

Total: 0.5 + 1 + 0.5 + 0 = 2/5.

Q.3(iv) OR — Power, Efficiency, and Impossibility of 100% Efficiency

5/5

Power defined as $P = W/t$ with unit Watt. Efficiency formula (% efficiency = useful output / total input $\times 100$) correctly stated. Numerical calculations: efficiency = $240/800 \times 100 = 30\%$; useful output power = $30\% \times 800 = 240$ W. Explanation of why 100% efficiency is impossible (some energy always converts to non-useful forms like heat and sound due to friction and air resistance) — complete and precise. Perfect score on this question.

Summary

Strengths

- Perfect MCQ score — 6th consecutive 12/12 (exams 14-19); Physics MCQ 60/60 all-time
- All three Newton's laws with mathematical notation including $F_{as} = -F_{sa}$
- Torque, work/energy, power/efficiency calculations all correct
- $P = \rho gh$ derivation fully shown step by step
- Q.3(iv)OR: power, efficiency calculation (30%), useful power (240 W), and impossibility of 100% efficiency — all 5/5
- Rotatory motion — two examples given (Earth's axis, fan blades)
- Translatory/rectilinear — correct example (car on straight road)
- Hooke's law formula, spring constant, and units — fully correct

Areas for Revision

- Vibratory motion: definition known, daily-life example omitted (pendulum, guitar string, tuning fork)
- Limitations of Newton's laws: not applicable at relativistic speeds or sub-atomic scales — must memorise
- Elasticity definition: "elastic limit" confused with "inelastic"; clarify the distinction
- Force-extension graph: MUST draw the actual curve, not just describe it
- Hooke's law practical applications: spring balance, watch balance wheel, galvanometer — 2-mark sub-part left entirely blank

Critical Pattern — Stopping Before All Sub-Parts (4th Consecutive Exam)

- This exam: Q.3(i) missing vibratory example, Q.3(ii) limitations absent, Q.3(iii) applications (2 marks) and sketch entirely missing. Combined cost: 4 marks in Section C.
- Previous exams: Physics-04 (2 marks), Bio-07 (3 marks). Total cost across last 3 exams from this pattern alone: ~9 marks.
- Drill fix: after completing each long answer, re-read the full question text and explicitly tick every named requirement before turning the page.

Physics Attempts — Comparison

#	Date	Exam	MCQ	Sec B	Sec C	Total	%	Grade
1	7 Apr	exam-01-ch1-2	12/12	24.5/33	27/20*	63.5/65	97.7%	A+
2	10 Apr	exam-02-ch3-4-9	12/12	—	—	62/65	95.4%	A+
3	18 May	exam-03-ch5	12/12	—	—	65/65	100%	A+
4	22 Jun	exam-04-ch6	12/12	31/33	12.5/20	55.5/65	85.4%	A
5	25 Jun	exam-07-ch1-6-9	12/12	31.5/33	15/20	58.5/65	90.0%	A+

* exam-01/02/03 section splits reconstructed from totals where individual section records were not retained.

Recovery from exam-04's 85.4% low. Section B improved from 31/33 to 31.5/33. Section C rose from 12.5/20 to 15/20. The persistent stopping-early pattern in Section C is the only barrier to returning to the 95-100% band.